

Research proposal.

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Topic: The effect of English-Spanish language in mathematical reasoning and ability.

Thesis question: Does language affect mathematical skills?

Formatting notes

In contrast to the description asked by the guidelines of the assignment, I have intentionally formatted the list of sources by type of source and whether they are a “yes” or a “no” to the thesis question, this was taken in the administrative essence of permitting the reader to scheme through the document and the sources with the capacity of identifying the position of the sources without a need to search outside of the document. With this in mind, the last bullet point of the model given, which states whether or not the source accepts or rejects the thesis question is ignored in writing (as that context was given in formatting).

Popular sources

Sources against the thesis question.

1. [1] Jude, “Does language restrict your thoughts?” [Online]. Available: <https://www.youtube.com/watch?v=7p16NS-M0TQ>
 - YouTube video. The platform serves for several opinion-vlogs of both scholarly and non-scholarly nature to be made, but it is generally considered as a popular source by the academic community due to the verbal explanation of some topics lacking the depth given by their referred sources, or that the sources for claims may completely be undisclosed or brushed upon lightly.
 - Jude (whose full name is undisclosed through the video) is a philosophy/linguistics content creator who argues that language determinism doesn’t fully justify the objectivity of mathematical-logic ideas. That is, our brains translate the ideas of mathematics from image and abstract components into language, and as such, the Sapir-Whorf hypothesis is rendered unusable to claim that language affects cognition in these topics.
 - Three quotes:
 1. “If we rely on the left [hemisphere of the brain’s] linguistic precision, we might indeed find some ideas inconceivable because they don’t fit into the neat categories and syntax of language.”
 2. “Einstein famously claimed that his ideas came in images and feelings, only later translated into words.”
 3. “[Steven] Pinker suggest that thoughts exist independent of language and are represented in a non-linguistic format...”
 - References the theory that the human brain forms thoughts, and abstract concepts unrelated to language in a way that ideas and mathematical thoughts (based on Einstein’s example) are separated from the brain’s analysis of language. Rejecting that language determinism or language universality are true in the realm of cognition. None of the claims are done informally or colloquially, but they go unsourced with the exception of major theories like the “mentalese” (Steven Pinker’s idea described above).
2. [2] National Council of Teachers of Mathematics, “Math Education Aspects of Students Who Use Two Languages.” [Online]. Available: <https://www.nctm.org/Research-and-Advocacy/Research-Brief-and-Clips/Math-Education-Aspects-of-Students-Who-Use-Two-Languages/>

- Journal's article. The reason why this source was included as popular is given by the criticism of the scientific community towards this specific journal, with mathematics and educators questioning the sourcing of material, the experimenting done, and the credibility of some of the arguments.
- The article claims that language isn't a blocker for mathematical intuition as it doesn't alter the quality of conceptual thinking. Albeit showing that there are differences in the arithmetic performance of adults using their non-preferred language.
- Three quotes:
 1. "Differences between monolingual and bilingual adults ranged from 0.2 seconds to 0.5 seconds (for average response times ranging between 2 and 3 seconds). Such differences seem negligible for interactions in classrooms. The differences in response time can be easily eliminated if bilingual speakers are permitted to use only one of their languages (McClain and Huang 1982). This last finding supports classroom practices that allow bilingual students to choose the language they use for arithmetic computation in the classroom."
 2. "We should not assume that bilingual students switch into their first language only because they are missing mathematical vocabulary. Although some bilingual learners may sometimes use their first language in this way, code-switching can serve many functions. Bilingual students may code-switch for politeness, humor, and other functions that are not necessarily related to either vocabulary or mathematics."
 3. "Studies focusing on computation say little regarding language switching during conceptual mathematical reasoning."
- Speaks with a personal voice, giving examples from their own experiences which is usually unlikely for purely academic - research based articles. Supports that code-switching (changing your language of speech) during mathematical instruction isn't a sign of weakness in knowledge in terminology, but it doesn't claim that there is a specific affection of code switching.
- 3. [3] r/math Community, "Non-English speakers: when doing math, do you think in your native language or English?" [Online]. Available: https://www.reddit.com/r/math/comments/1n4ztdl/nonenglish_speakers_when_doing_math_do_you_think/
 - Post in Reddit, a popular social media site utilized for blog posts in distinct communities. In this case, the surveyed community is mostly comprised of professional and amateur mathematicians of the "r/math" community.
 - The diversity of contributors in this post show a split in opinions, with many opinions making claim that they think in math utilizing their native languages, some others saying they utilize english, and many more claiming that they use first order logic (language of math utilized for the evaluation of logical propositions which can only be evaluated true or false, there is a split on whether one can think in first order logic without the need to parse to a language), and some other claim to think in "just math".
 - Three quotes [responding to the question: Non-english speakers: when doing math, do you think in your native language or english?]:
 1. "Mostly in language-free ideas. Otherwise I think my brain jumps between both languages as it sees fit "
 2. "I think exclusively in first-order logic with the addition of single non-logical binary predicate. [i.e the author of the comment claims to think in first order logic, adjoined with one non logical (common language, second order, cannot be true or false) sentence, without specifying language.]"

3. "I like to think in shapes and drawings and with voice over mixed with languages I knows [know]. "
- Informal language, a lot of satire corresponding to the mathematical community's humor over this topic, and there is a lack of accountability of which accounts correspond to professional mathematicians and which don't.

Sources in pro of the thesis question.

1. [4] Language Lab, "The Fascinating World of Language and Math in Bilingual Minds." [Online]. Available: <https://medium.com/language-lab/the-fascinating-world-of-language-and-math-in-bilingual-minds-b081289d77fd>
 - Post in Medium.com, a site utilized in the simplification of mathematics and computer science tutorials, and STEM related blog posting. There is no need for credentials in Medium.com, nor is there a system that filters out or increases the validity of the content within all posts in the site.
 - The webpost makes a claim that the Polish speaking author "code-switches" to Polish (their mother language) at the moment of doing math.
 - Three quotes:
 1. "The parts of the brain that handle language are mostly in the left hemisphere for right-handed people, and they are close to the parts that handle math. Even though they live close to each other, that doesn't mean they speak the same language."
 2. "The way you see and understand the world?" Yes, it is. My brain, on the other hand, goes back to its old ways when it comes to math. When someone is angry or scared, they might switch back to their original language. But in this case, it's not anger or fear that wakes up the mother tongue; it's the beautiful ballet of algebra."
 3. "Why, then, stick to Polish for numbers when English rules every other area of thought? The idea of cognitive fluency and emotional connection holds the key to the answer. When someone learns their first math, it often gets ingrained in their brain in the same way that their first language does. This basic education sets up a default language model for mathematical thinking. For this example, Polish isn't just a language; it's a cognitive scaffold that helps with mathematical thought by making it easier to understand numbers."
 - The entirety of the article goes uncited in terms of research, is spoken in terms of personal experiences, and attempts to appeal to the reader's feelings about the topic through mental imagery like "ballet of algebra".
2. [5] El Recreo Spanish, "Bilingual Math: How Spanish Boosts Your Child's Brain." [Online]. Available: <https://www.elrecreospanish.com/blog/bilingual-math-spanish-cognitive-benefits-children>
 - El recreo is a website on children's education, written in the form of an editorial article and often times lacking of formality and sources. Furthermore, the site is motivated by the promotion of their originating franchise's Spanish classes.
 - The website makes the claim that Spanish boosts children mathematical capacities, specifically doubling the brain pathways per language used, augmenting cognitive executive function.
 - Three quotes
 1. "The Spanish number system is highly logical and transparent. For example, "sixteen" in English gives no clue about its composition, while "dieciséis" in Spanish clearly means "ten and six." This linguistic clarity helps children understand place value concepts earlier."
 2. "Parents who introduce Spanish math learning report impressive developments in their children:"

3. "Switching between Spanish and English requires mental control to suppress one language while activating another. This constant practice strengthens attention control mechanisms in the brain, helping children focus better during all learning activities."
- The tone of the website is the one of a promotion, more closely related what you would write in a pamphlet for a school about to start their registration season. There are no sources cited.
3. [6] S. D. Sparks, "How Language Development Can Boost English Learners' Math Skills." [Online]. Available: <https://www.edweek.org/teaching-learning/encouraging-mental-flexibility-can-help-boost-english-learners-math-skills/2025/06>
- Written as an editorial webpost by S.D. Sparks, who writes about pedagogy in general. The website, EdWeek proposes to be a website on news and articles in education.
 - The site makes the claim that english learners may benefit from the development of language skills, as multilingual students outperform in executive function the monolingual students. The rest of this article answers question on how to resolve the disparity in language skills for math in the classroom. Explained in the form of an interview.
 - Three quotes:
 1. "Multilingual students often outperform monolingual students in the executive functioning skills needed to start solving a problem in one way and then shift to a different strategy in response to changing demands. "
 2. "A lot of times we tend to think that because students speak Spanish, they've all got the same education, [when] it's important to understand the different backgrounds that students bring, and not automatically assume that because the students come from a respective country, that they automatically have this rich math education or that they don't. There have been a lot of cases where [newcomer] students have been more advanced than some of my [native-English-speaking] students that I have in the classroom."
 3. "It's important to gauge what students know about a topic. If we're doing equations, what do we know about "equation," and what is an expression? We can flush out some of those semantics and even some of those misconceptions that might come up later on in the class."
 - The website is complete in the sense of its content as an interview being reflective and accurate of the opinions of the subject interviewed (Pedro Rodriguez, a math and science dual-language teacher at the K-8 P.S. 161 Pedro Albizu Campos in Manhattan), there isn't much formal information on the topic but there is the implication that learning changes culturally.

Formal sources

It is to be noted that this topic has never been of matter to the supreme court, or any institution of law in the United States whatsoever. This is because the thesis question is too naive in terms of politics and may be brushed over without making a case that would require an executive decision in the realm of law. To resolve the lack of sources, I will introduce as formal sources one paper in Spanish, and another in english which cover Homology groups.

A simplicial complex is a chain of points connected together by triangles, or their equivalent in more dimensions. By studying the way these complexes map into a shape, we are left with a device to do algebraic works over the shape's boundaries: a p -chain group (for p dimensions), notated $C_p(X)$ when X is a space (just any shape with definition of what regions are close to each other).

Roughly speaking, the $C_p(X)$ group as an algebraic device, permits the user to define the homology group $H_p(X)$, which is relevant in the study of algebraic geometry. If you don't believe me, we can study the claim "A donut and a cup of coffee are mathematically equivalent" from the $H_p(X)$ group.

Namely, a number notated as β_r , called the Betti number is calculated from the “rank” of a homology group. The least number of elements a smaller group needs to have to generate the entire group.

Although I make the same genre of analysis on both of these papers as the ones presented before, none claim a position over the thesis question, and instead they are meant to be compared as grounded sources of information which can be comparatively analyzed.

1. [7] P. Nadathur, “An Introduction to Homology.” 2007.

- Article published online by the University of Chicago. A reputable university which commonly peer reviews content before publication to any of its stemming websites.
- The article itself only explains an introduction to homology groups. That is, there isn’t any relevant opinion expressed in the topic.
- Three quotes:
 1. “Given the set S of vertices of a simplex, we define an orientation on the simplex by selecting some particular ordering of S . Vertex orderings that differ from this by an odd permutation are then designated as reversed, while even permutations are regarded as unchanged. Any simplex, then, has only two possible orientation”
 2. “Path connected $\Leftrightarrow$ Connected”
 3. “The boundary map $\delta_n(\sigma) = \sum_i (-1)^i \sigma|_{[v_0 v_1 \dots v_{i-1} v_{i+1} \dots v_n]}$ where v_i are the 0 –simplices of σ , that is, the maps of the vertices of $\Delta^n : v_i : \Delta_0 \longrightarrow X$ ”
- It has a formal tone, cited sources, and the mathematical statements are formatted as formally done in the field. Unlike the paper in spanish, it lacks illustration and it instead works with the graphics of mappings, it is also much more emphasized to pure algebra.

2. [8] W. M. Villate and S. Huérfano, “Introducción a la Homología.” Bogotá D.C., Colombia, 2005.

- Published in Academia.edu, a popular source in academia given its nature of being more of a social network for researchers than a journal or publisher, despite of having content posted in the website.
- The article explains the same topic.
- Three quotes:
 1. “Un 2-simplejo ($P_0 P_2 P_1$) es un triángulo con su interior, considerado con la orientación contraria a la de las manecillas del reloj. [Example of a 2 –simplex]”
 2. “Consideremos el toro T^2 . La superficie del toro no es frontera de ninguna 3-cadena, es decir, $B_2(T^2) = 0$. Esto implica que $H_2(T^2)$ es generado, por un solo generador: la superficie misma, por tanto $H_2(T^2) \cong \mathbb{Z}$ [On how the 2 dimensional homology of the 2-dimensional torus (annulus) is equivalent to the integers ...-1,0,1,2,3,... additively].”
 3. “los grupos de homología son invariantes topológicos. [Homology groups are topologically invariant, meaning that if a shape is deformed, the group would still be the same.]”
- It has a formal tone, cited sources, and once again, the mathematical structure as used commonly in the field is there. The terminology in spanish has adapted the language from english, for example Simplex $\rightarrow$ Simplejo, Torus $\rightarrow$ Toro. This last case being interesting because “toro” in Spanish is the word for bull. Although the etymology for “torus” is the same in both languages.

Bibliography

[1] Jude, “Does language restrict your thoughts?” [Online]. Available: <https://www.youtube.com/watch?v=7p16NS-M0TQ>

- [2] National Council of Teachers of Mathematics, “Math Education Aspects of Students Who Use Two Languages.” [Online]. Available: <https://www.nctm.org/Research-and-Advocacy/Research-Brief-and-Clips/Math-Education-Aspects-of-Students-Who-Use-Two-Languages/>
- [3] r/math Community, “Non-English speakers: when doing math, do you think in your native language or English?” [Online]. Available: https://www.reddit.com/r/math/comments/1n4ztdl/nonenglish_speakers_when_doing_math_do_you_think/
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